

RAILWAY TICKET BOOKING SYSTEM

School of Computing

**Bachelor of Technology
in
Computer Science & Engineering**

By

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Full Stack Application Development

Slot : E1



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May, 2026

BONAFIDE CERTIFICATE

It is certified that the work contained in the Full Stack Application Development report titled "RAILWAY TICKET BOOKING SYSTEM" by Penumalli Midhilesh Reddy (VTU25418) has been carried out in Winter semester of Academic year 2025-2026(WS2526).

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DECLARATION

I declare that this written submission represents my ideas in my own words and where others' ideas or words have been included, we have adequately cited and referenced the original sources. I also declare that i have adhered to all principles of academic honesty and integrity and have not misrepresented or fabricated or falsified any idea/data/fact/source in our submission. I understand that any violation of the above will be cause for disciplinary action by the Institute and can also evoke penal action from the sources which have thus not been properly cited or from whom proper permission has not been taken when needed.

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ABSTRACT

The Railway Ticket Booking System is a web-based application developed to automate and simplify the process of booking train tickets. Traditional ticket booking methods involve long queues, manual effort, and lack of real-time updates, making the process inefficient and time-consuming.

This system allows users to search for trains, check seat availability, and book tickets online. The application provides a user-friendly interface where users can enter journey details such as source, destination, and travel date to view available trains. It also enables users to enter passenger details and confirm bookings easily.

The system ensures real-time updating of seat availability and maintains data consistency using a backend database. It minimizes human errors and improves efficiency. The project is developed using modern technologies such as React.js for frontend, Spring Boot for backend, and MySQL for database management.

Keywords: Full Stack Web Development, Campus Recruitment System, React.js, Node.js, MySQL, JWT Authentication, Recruitment Analytics

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LIST OF ACRONYMS AND ABBREVIATIONS

API	Application Programming Interface
DBMS	Database Management System
IRCTC	Indian Railway Catering and Tourism Corporation
OTP	One-Time Password
PNR	Passenger Name Record
UI	User Interface

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Chapter 1

INTRODUCTION

1.1 Introduction

The Railway Ticket Booking System is a modern full stack web application developed to improve and digitalize the train ticket reservation process. Traditional railway booking systems often depend on manual operations and disconnected processes, making ticket booking difficult and inefficient. The proposed system provides a centralized platform where users can search for trains, check seat availability, book tickets, track bookings, and manage their details through an interactive dashboard. Administrators can manage trains, schedules, seat availability, bookings, reports, and overall system activities efficiently. The platform is developed using React.js, Spring Boot, and MySQL with secure authentication and responsive user interfaces to provide a reliable and scalable railway booking environment.

1.2 Aim of the Project

The main aim of the project is to develop a secure and modern railway ticket booking system that simplifies train reservation activities for both users and administrators. The system aims to provide centralized train management, secure authentication, seat availability tracking, booking management, and responsive user interfaces to improve the efficiency of the ticket booking process.

1.3 Scope of the Project

The scope of the project includes user registration, secure login systems, train management, ticket booking management, seat availability tracking, booking history, notifications, and dashboard systems. The platform supports both user and admin modules and can further be enhanced with mobile applications, online payment integration, cloud deployment, AI-based recommendations, and advanced analytics systems in the future.

1.4 Framework

The Railway Ticket Booking System follows a full stack web application architecture consisting of frontend, backend, and database layers. The frontend is developed using React.js and CSS to provide

responsive and interactive user interfaces. The backend is developed using Spring Boot to manage APIs, authentication, and business logic. MySQL is used as the relational database for storing users, trains, bookings, and ticket details. Secure authentication and password encryption are implemented to ensure safe access and data protection within the system [Figure 1.1].

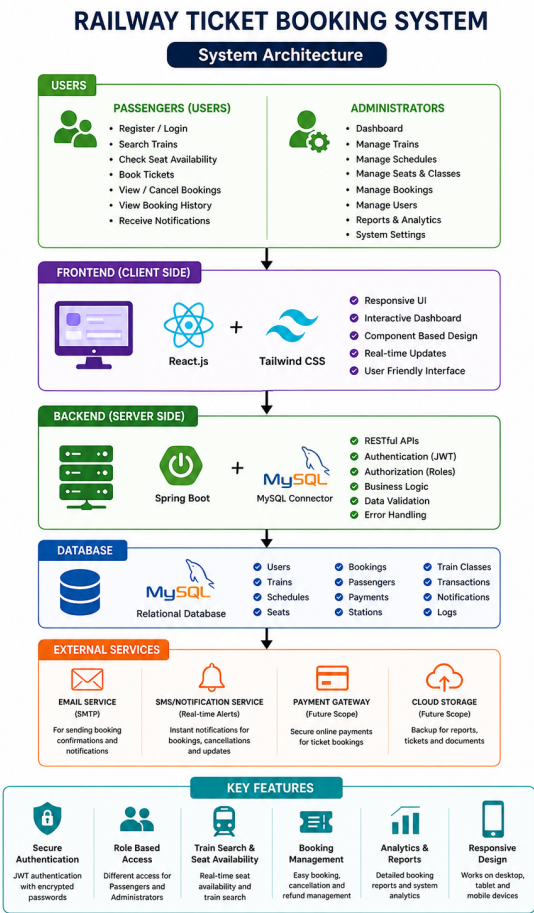


Figure 1.1: System Framework Architecture

Figure 1.1: System Frame Architecture

Chapter 2

SURVEY OF SIMILAR APPLICATION IN MARKET

The rapid growth of digital technologies has led to the development of several online ticket booking platforms that simplify travel and reservation activities. Many railway systems and transportation services use web-based booking systems to manage train schedules, seat availability, and passenger reservations efficiently. Existing platforms such as IRCTC and other online railway booking portals provide ticket booking, seat availability checking, and journey management services for passengers.

IRCTC is one of the most widely used railway booking platforms that allows users to search trains, check seat availability, and book tickets online. However, it often faces issues such as heavy traffic, slow performance during peak hours, and limited user experience improvements.

Many existing railway booking systems focus mainly on ticket reservations but lack advanced features such as intuitive dashboards, enhanced user interfaces, and efficient booking management systems. Some platforms also have complex navigation, making it difficult for users to complete bookings .

Traditional railway booking systems still rely on manual counters and basic digital systems with limited functionality. These systems often suffer from poor user interface design, lack of real-time updates, and minimal security features. They also do not provide centralized analytics, booking tracking, or efficient management tools for administrators.

The proposed Railway Ticket Booking System overcomes these limitations by providing a centralized and modern booking platform with secure authentication, responsive user interfaces, real-time seat availability tracking, booking management, notifications, and role-based access control. The system combines essential booking features with improved usability and efficient management capabilities to provide a reliable and scalable solution.

Chapter 3

FRAMEWORK DESCRIPTION

3.1 Frontend Implementation

3.1.1 Environment Setup

The frontend of the Railway Ticket Booking System was developed using React.js and CSS to provide a responsive and user-friendly interface. The development environment was configured using Node.js, npm, and Vite. Required frontend dependencies such as React Router DOM and Axios were installed to support routing and API integration. Visual Studio Code was used as the primary development environment for implementing and managing the frontend architecture.

3.1.2 Structure of the Project

The frontend project follows a component-based architecture where the application is divided into reusable components, pages, layouts, and utility modules. The project structure contains separate folders for authentication pages, train search pages, booking pages, dashboards,

reusable UI components, API services, and routing configuration. This modular structure improves maintainability, scalability, and code reusability within the system.

3.1.3 Execution Procedure

The frontend application is executed using the Vite development server. After installing the required dependencies using `npm install`, the application is started using the `npm run dev` command. The frontend communicates with backend APIs through Axios requests and renders dynamic content such as train listings, seat availability, booking details, notifications, and user dashboards.

3.2 Backend Implementation

3.2.1 Environment Setup

The backend of the Railway Ticket Booking System was developed using Spring Boot and Java. The backend environment was configured using required dependencies such as Spring Web, Spring Data JPA, MySQL Driver, and Spring Security. The backend APIs were designed to handle authentication, train management, booking management, seat availability, notifications, and database operations securely and efficiently.

3.2.2 Structure of the Project

The backend project follows a layered architecture consisting of controllers, services, repositories, configuration files, and model classes. API modules are implemented for authentication, user operations, admin operations, train management, and booking management. Security configurations are used for authentication, authorization, validation, and error handling to ensure secure and scalable backend operations.

3.2.3 Execution Procedure The backend server is executed using Spring Boot. After configuring the project and dependencies, the application is started using the Run option or `mvn spring-boot:run` command. The backend establishes a secure connection with the MySQL database and exposes RESTful APIs for frontend communication. Authentication, CRUD operations, seat availability updates, and booking management are processed through the backend server [Figure 3.1].

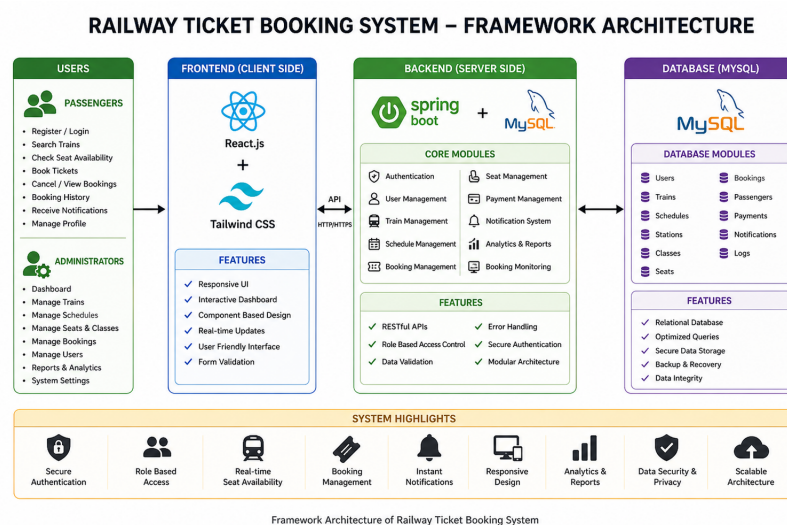


Figure 3.1: Frontend and Backend Framework Architecture

3.3 Database Implementation

3.3.1 Database Configuration

The Railway Ticket Booking System uses MySQL as the relational database management system for storing application data securely and efficiently. The database was configured locally using MySQL Workbench. Environment variables were used to establish secure database connections between the backend server and the MySQL database.

3.1.2 Schema Structure

The Railway Ticket Booking System uses MySQL as the relational database management system for storing application data securely and efficiently. The database was configured locally using MySQL Workbench. Environment variables were used to establish secure database connections between the backend server and the MySQL database.

3.1.3 Tables created

The following tables were created in the MySQL database for the Railway Ticket Booking System.

Chapter 4

METHODOLOGIES

4.1 Project Architecture

The Railway Ticket Booking System follows a three-tier architecture using React.js, Spring Boot, and MySQL to provide secure, scalable, and efficient ticket booking and management services [Figure 4.1].

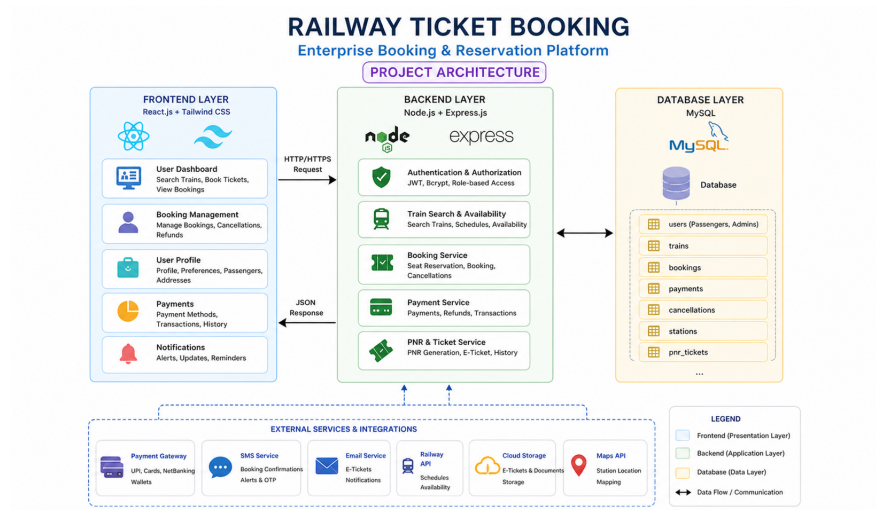


Figure 4.1: Project Architecture

4.2 DataFlow Diagram

The data flow within the Railway Ticket Booking System begins with users and administrators interacting with the frontend interface. User requests are sent to the backend server through RESTful APIs. The backend processes authentication, validates requests, manages business logic, and communicates with the MySQL database to store or retrieve information. The processed data is then returned to the frontend and displayed dynamically through dashboards, train search pages, booking modules, and notification systems. This structured data flow ensures smooth communication between all components of the system.

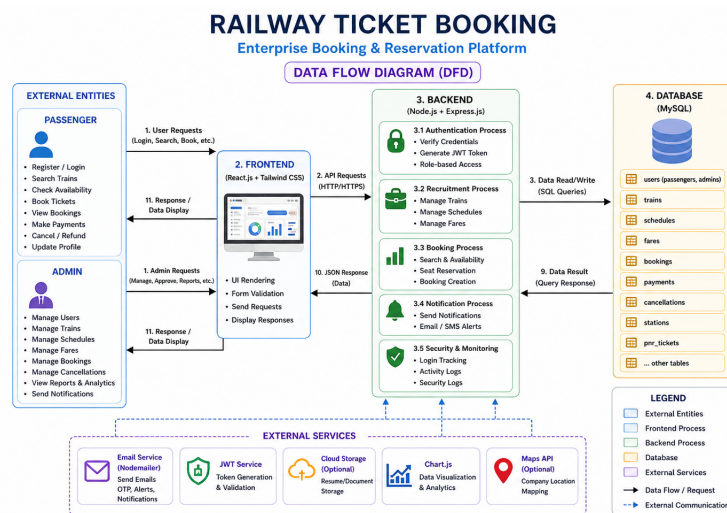


Figure 4.2: DataFlow Diagram

4.3 Module 1:Authentication Module

The authentication module manages secure login and signup functionality for both users and administrators. JWT-based authentication and password encryption are implemented to ensure secure access management [Figure 4.3]. Users are required to register before accessing the system, and role-based authentication is used to protect dashboards and APIs from unauthorized access.

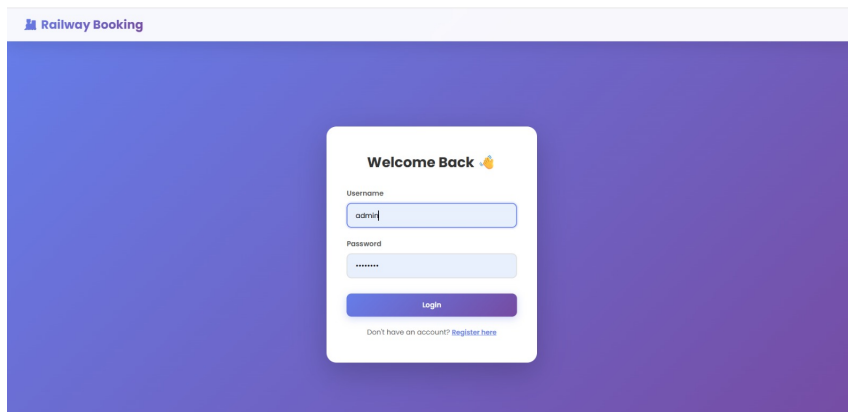


Figure 4.3: Authentication

4.4 Module 2:Ticket Booking Management Module

The ticket booking management module handles train management, ticket bookings, seat availability tracking, and passenger booking activities. Administrators can add trains, manage schedules, update seat availability, and monitor booking records. Users can search trains, check availability, book tickets, and track their booking status through

the dashboard [Figure 4.4].

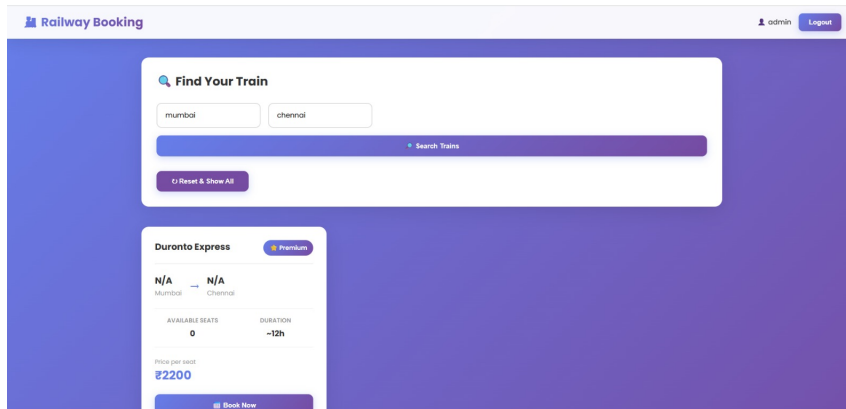


Figure 4.4: Ticket Booking Management

4.5 Module 3: Analytics and Monitoring Module

The analytics and monitoring module provides dashboard analytics, reports, notifications, booking tracking, and system monitoring features. Interactive charts and graphical reports are generated to visualize booking statistics, seat availability, user activity, and overall system performance [Figure 4.5].

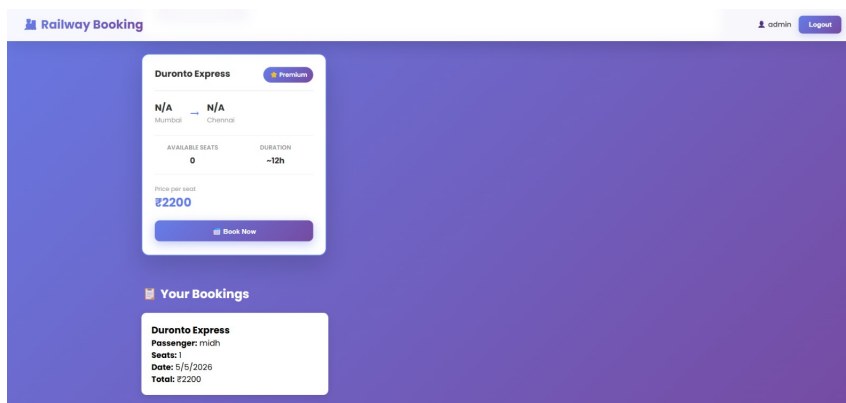


Figure 4.5: Analytics and Monitoring

Advantages of this Project

a) Secure Authentication: Secure authentication mechanisms ensure protected login systems and controlled access for both users and administrators.

b) Centralized Booking Management: The system allows centralized management of trains, schedules, bookings, seat availability, notifications, and reports through a single integrated platform.

c) Real-Time Updates and Dashboard Systems: Interactive dashboards and real-time updates provide better monitoring of booking activities, seat availability, and system performance.

Disadvantages

The system requires continuous internet connectivity for accessing online booking features and services. Initial setup and deployment may require technical knowledge related to full stack development, database configuration, and server management. Large-scale deployment may also require cloud infrastructure and advanced security optimization for handling high user traffic efficiently.

Chapter 5

RESULTS AND DISCUSSIONS

The Railway Ticket Booking System was successfully designed and implemented using React.js, Spring Boot, and MySQL. The system provides secure authentication, train management, ticket booking, seat availability tracking, dashboard systems, notifications, and booking monitoring functionalities. The frontend interface was developed with responsive UI components and modern layouts to provide an interactive user experience for both users and administrators.

The User Dashboard allows passengers to search trains, check seat availability, book tickets, view booking history, and manage their profiles efficiently [Table 7.2]. The Admin Dashboard provides centralized management of trains, schedules, bookings, seat availability, reports, and overall system activities.

The MySQL database successfully stores and manages users, trains, bookings, ticket details, notifications, and system data [Table 5.1]. Secure authentication mechanisms ensure protected login functionality

and controlled access across the system.

The project demonstrates scalable architecture, responsive design, secure authentication, and efficient ticket booking management suitable for real-world railway reservation systems. Content with screenshots

Table 5.1: Railway Ticket Booking System - Feature Comparison

Module / Feature	Description / Advantage
Login & JWT Authentication	Secure user authentication using JWT tokens to ensure protected access for users and admins.
Seat Availability Tracking	Real-time seat availability updates after every booking to avoid overbooking issues.
Payment Gateway Integration	Secure and fast online payment processing for ticket booking transactions.
Ticket PDF Generation	Automatically generates downloadable and printable PDF tickets after successful booking.
IRCTC-like UI Design	User-friendly interface designed similar to IRCTC for better user experience and easy navigation.

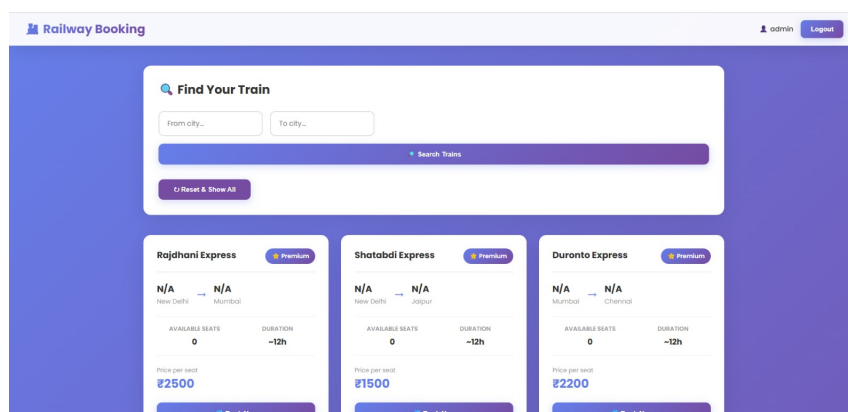


Figure 5.1: FINAL PROJECT

Chapter 6

CONCLUSION AND FUTURE ENHANCEMENTS

6.1 Conclusion

The Railway Ticket Booking System was successfully designed and developed as a modern full stack ticket reservation system. The platform provides a centralized environment for managing train schedules, ticket bookings, seat availability, user activities, and dashboard systems. The project integrates secure authentication, responsive user interfaces, booking management, notifications, and role-based access control to improve the efficiency and reliability of the booking process.

The User Dashboard enables passengers to search trains, check availability, book tickets, track bookings, and manage their profiles through an interactive and user-friendly interface. The Admin Dashboard provides centralized control for managing trains, schedules, bookings, reports, and system monitoring features. The use of React.js, Spring

Boot, and MySQL ensures scalability, maintainability, and efficient database operations within the system.

The project demonstrates modern web development concepts, secure authentication mechanisms, responsive dashboard systems, and efficient ticket booking management suitable for real-world railway reservation environments.

6.2 Future Enhancements

The Railway Ticket Booking System can be further enhanced with several advanced features to improve scalability and user experience. Future improvements may include online payment integration, mobile application support, cloud-based deployment, real-time notifications, and advanced analytics dashboards. Additional features such as ticket PDF generation, QR code-based ticket verification, chatbot assistance, and smart seat recommendation systems can also be integrated into the system to provide a more advanced booking experience. The system can also be expanded using cloud technologies and scalable architectures to support large-scale deployment and handle high user traffic efficiently.

Chapter 7

ADDRESSING COMPLEX ENGINEERING PROBLEM AND SDG

7.1 Complex Engineering Problem (CEP)

Table 7.1: Mapping of Project to Complex Engineering Problem (CEP)

Level	Attribute	Justification
WP1	Depth of Knowledge	Requires knowledge in Spring Boot, MVC architecture, database design, REST APIs, and frontend technologies (Thymeleaf, HTML/CSS) for developing a scalable event management system.
WP2	Conflicting Requirements	Balances usability and security by ensuring easy student access while maintaining secure admin authentication and data validation.
WP3	Depth of Analysis	Involves system design, database schema creation, validation handling, and efficient event filtering and registration mechanisms.
WP4	Familiarity	Integrating real-time event updates, user interaction, and admin analytics introduces challenges beyond traditional manual systems.
WP5	Applicable Codes	Requires custom implementation for event filtering, registration tracking, and statistics generation not fully covered by standard templates.
WP6	Stakeholder Needs	Addresses requirements of students, faculty coordinators, and administrators ensuring smooth event organization and participation.
WP7	Interdependence	Integrates frontend UI, backend services, database, validation, and security modules into a unified system.

7.2 Complex Engineering Activities (CEA)

Table 7.2: Mapping of Project to Complex Engineering Activities (CEA)

EA No.	Attribute	Justification
EA1	Range of Resources	Utilizes Spring Boot, Spring MVC, Spring Data JPA, Thymeleaf, HTML/CSS, and relational databases for full-stack development.
EA2	Level of Interactions	Manages interactions between event management, user registration, validation, authentication, and reporting modules.
EA3	Innovation	Implements a web-based centralized platform with filtering, real-time updates, and secure admin operations.
EA4	Consequences to Society and Environment	Enhances student engagement and improves institutional efficiency while reducing manual errors and delays.
EA5	Familiarity	Combines web development, security, and data analytics into a unified system, extending beyond traditional event handling methods.

7.3 SDG Mapping (CEP Section)

Table 7.3: SDG Mapping for the Project

SDG No.	SDG Title	Relevance to the Project
SDG 4	Quality Education	Facilitates student participation in academic events, workshops, and seminars, enhancing learning beyond the classroom.
SDG 9	Industry, Innovation and Infrastructure	Promotes digital transformation in educational institutions through a centralized event management system.
SDG 11	Sustainable Cities and Communities	Supports smart campus initiatives by digitizing event processes and improving resource management.
SDG 16	Peace, Justice and Strong Institutions	Ensures secure and transparent event handling through authentication and controlled access.
SDG 12	Responsible Consumption and Production	Reduces paper usage by enabling online event registration and management.

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